

Ahmed Raza

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github.com/ahmedrazaposwal
Kaiserslautern, Germany (Willing to Relocate)



PROFILE

Software Engineer with 3+ years of experience in Java, Spring Boot, REST APIs, PostgreSQL, and cloud-native backend development. Worked at Fraunhofer Gesellschaft in Germany, developed scalable backend systems and data-driven applications. Experienced in microservices, CI/CD, Docker, cloud technologies, and AI-based systems. Seeking Backend Engineer / Java Developer / Software Engineer roles in Germany.

EXPERIENCE

Fraunhofer Gesellschaft

Software Developer

Sep 2022 – Apr 2026

Kaiserslautern, Germany

- Designed and maintained Spring Boot microservices and REST APIs used in research and industrial analytics applications at Fraunhofer ITWM.
- Optimized PostgreSQL queries and backend services, reducing API response times by 25% and improving system scalability.
- Implemented CI/CD pipelines using GitHub Actions and Docker, reducing deployment effort and improving release reliability.
- Applied test-driven development using JUnit 5, increasing test coverage to more than 80%.
- Resolved more than 50 software defects across backend applications, improving stability and reducing user-reported issues.
- Developed data visualization and reporting tools using Python Streamlit and R Shiny, enabling researchers to analyze large datasets efficiently.
- **Technologies:** Java, Spring Boot, Spring Security, REST APIs, Microservices, PostgreSQL, JPA/Hibernate, JUnit 5, Docker, GitHub Actions, CI/CD, Python, Streamlit, R Shiny

RRLAB, TU Kaiserslautern

Master Thesis Researcher – LLM Autonomous Vehicle Control

Nov 2024 – Sep 2025

Kaiserslautern, Germany

- Developed a C++ framework for natural-language control of autonomous off-road robots.
- Integrated large language models, retrieval-augmented generation, and computer vision into a unified AI control pipeline.
- Used LLaMA and Gemini models to support reasoning and task execution for robotic control scenarios.
- Tested the system across more than 25 scenarios and validated the approach on physical robotic hardware.
- **Technologies:** C++, Finroc, Python, LLMs, RAG, Computer Vision, JSON, Git, NVIDIA Jetson

Codecture

Android Application Developer

Aug 2019 – Jan 2020

Pakistan

- Developed Android applications in Java, covering backend integration, user interface implementation, and application logic.
- Integrated REST APIs and JSON-based communication for reliable real-time data synchronization.
- Designed user-friendly interfaces following Material Design principles.
- Performed debugging, testing, and maintenance to improve application stability and usability.
- Worked in an agile Scrum team with regular bi-weekly release cycles.
- **Technologies:** Java, Android Studio, MySQL, REST APIs, JSON, Agile, Scrum

PROJECTS

ERP Management System

- Developed a Spring Boot ERP system with JWT-based authentication and role-based access control.
- Designed backend modules using the Controller–Service–Repository architecture pattern.
- Implemented product and inventory management features with CRUD operations, pagination, filtering, and soft delete.
- Integrated PostgreSQL with JPA/Hibernate and used DTOs for clean data transfer between layers.
- **Technologies:** Java, Spring Boot, Spring Security, JWT, PostgreSQL, JPA/Hibernate, REST APIs, Maven, DTOs

Smart Gardening System

- Developed an Android application for monitoring and controlling irrigation systems using real-time sensor data.
- Integrated Arduino hardware with Firebase Realtime Database for remote communication and control.
- Implemented automated irrigation scheduling to improve water efficiency and reduce manual effort.
- **Technologies:** Java, C++, Arduino, Firebase Realtime Database, Android, REST APIs

EDUCATION

Technische Universität Kaiserslautern <i>Master of Science (M.Sc.) in Informatics</i>	Apr 2022 – May 2026 <i>Kaiserslautern, Germany</i>
- Specialization: Software Engineering, Artificial Intelligence, and Machine Learning. - Master Thesis: Natural-Language Control of Autonomous Vehicles using LLMs, RAG, and Computer Vision.	
COMSATS University Islamabad <i>Bachelor of Science (B.Sc.) in Software Engineering</i>	Mar 2016 – Mar 2020 <i>Pakistan</i>
- Specialization: Software Engineering, Object-Oriented Programming, Database Systems, and Software Architecture.	

SKILLS

Backend: Java, Spring Boot, Spring Security, JPA/Hibernate, Microservices, REST APIs, JSON, JWT, OAuth2, Maven, Gradle
Databases: PostgreSQL, MySQL, MongoDB, Firebase
Cloud & DevOps: Docker, Kubernetes, AWS, GitHub Actions, CI/CD Pipelines
Testing: JUnit 5, Test-Driven Development, Unit Testing
Frontend & Mobile: React, Android, JavaScript, HTML5, CSS3
Programming Languages: Java, Python, C++, JavaScript, R
Tools & Methods: Git, Linux, Kafka, Redis, Jira, Agile, Scrum, IntelliJ IDEA, VS Code, GitHub Copilot, Claude

LANGUAGES

English – Proficient German – B1 Intermediate

CERTIFICATES

AWS Cloud Practitioner Essentials <i>Amazon Web Services</i>	May 2026
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